

Deep Learning

Image Generation

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1 Introduction

This project explores image generation using three fundamentally different families of generative models trained on the Cats dataset. We plan to systematically compare a generative adversarial network, a lightweight denoising diffusion probabilistic model, and a β -variational autoencoder in terms of image quality, training stability, and latent space structure. All three architectures are deliberately constrained in size to remain feasible within the available compute budget. Beyond core generation, we plan to investigate hyperparameter sensitivity, latent space interpolation, and extend the study with inpainting and style transfer tasks as well as a comparative cats-and-dogs experiment.

2 Methodology

2.1 Generative Architectures

We plan to implement and compare three generative models, each representing a different family:

DCGAN - a Deep Convolutional Generative Adversarial Network [1]. The generator maps a 100-dimensional latent vector $z \sim \mathcal{N}(0, I)$ through transposed convolutions with batch normalisation and ReLU activations, producing 64×64 RGB images. The discriminator mirrors the generator with strided convolutions, Leaky ReLU activations, and a sigmoid output. Training will follow the standard minimax objective with the Adam optimiser for both networks. This architecture will serve as the GAN representative.

Tiny DDPM - a lightweight Denoising Diffusion Probabilistic Model [2]. To reduce training and sampling time, it is planned to constrain the U-Net backbone: we will use a base channel width of 32, two resolution levels with a single residual block each, and no self-attention. The forward process will use $T = 200$ timesteps with a cosine variance schedule. The reverse process is parameterised by the noise prediction objective ϵ_θ . Sampling is performed with the standard DDPM ancestral sampler. Images are generated at 64×64 resolution. The total parameter count is estimated at approximately 2 million.

β -VAE - a β -Variational Autoencoder [4]. The architecture will be identical to a standard convolutional VAE: the encoder maps 64×64 inputs to a latent distribution $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x)I)$ with dimensionality $d = 64$, and the decoder reconstructs the image via transposed convolutions. The key modification is the introduction of a weight $\beta > 1$ on the KL divergence term in the training objective. The stronger KL penalty encourages individual latent dimensions to correspond to independent generative factors. This property can make the β -VAE interesting for the latent space traversal performance evaluation. We will use a fixed β throughout training.

2.2 Image Preprocessing

The Cats dataset contains images of varying resolution. We adopt the following preprocessing pipeline:

- All images are resized to 64×64 pixels using bilinear interpolation, chosen as a balance between visual quality and computational feasibility.
- Pixel values are normalised to $[-1, 1]$ for DCGAN and Tiny DDPM (matching the tanh output of the generator and the cosine noise schedule assumptions), and to $[0, 1]$ for the β -VAE.

2.3 Hyperparameter Investigation

For each model, we will investigate three hyperparameters while keeping others at their baseline values, following the same sequential tuning strategy as in previous projects.

2.3.1 DCGAN

1. **Learning rate:** we compare three values for both generator and discriminator: $\{10^{-5}, 10^{-4}, 10^{-3}\}$, with 10^{-4} as the baseline.
2. **Latent dimension $\dim(z)$:** we evaluate $\{64, 100, 256\}$ to assess the trade-off between expressiveness and mode coverage. The baseline is $\dim(z) = 100$.
3. **Batch size:** we compare $\{32, 64, 128\}$, with 64 as the baseline.

2.3.2 Tiny DDPM

1. **Number of diffusion timesteps T :** we compare $\{100, 200, 400\}$ to evaluate the trade-off between sample quality and sampling speed within the small-model regime. The baseline is $T = 200$.
2. **Base channel width:** we evaluate $\{16, 32, 64\}$ channels in the U-Net backbone to assess the effect of model capacity on generation quality under a fixed compute budget. The baseline is 32 channels.
3. **Learning rate:** we test $\{10^{-5}, 10^{-4}, 10^{-3}\}$, with 10^{-4} as the baseline.

2.3.3 β -VAE

1. **KL weight β :** we test $\{1, 4, 8\}$, where $\beta = 1$ corresponds to the standard VAE. The baseline is $\beta = 4$.
2. **Latent dimensionality d :** we evaluate $\{32, 64, 128\}$ to examine the effect on the disentanglement-capacity trade-off. The baseline is $d = 64$.
3. **Learning rate:** we test $\{10^{-4}, 10^{-3}, 10^{-2}\}$, with 10^{-3} as the baseline. VAEs should typically tolerate higher learning rates than GANs, however practical tests will verify this assumption.

2.4 Latent Space Interpolation

For each model, we select two generated images together with their latent vectors (or initial noise matrices in the case of Tiny DDPM). We linearly interpolate between the two latent representations to produce 8 additional vectors, use them to generate 8 new images, and present all 10 results (the 2 original and 8 interpolated). The outcomes will be discussed qualitatively, focusing on the smoothness of transitions and the overall coherence of intermediate images. The details of the planned evaluation metrics are described in the 3.2 section.

2.5 Inpainting

We plan to adapt the Tiny DDPM to perform image inpainting - the task of reconstructing a masked region of a real image. The procedure is as follows:

1. We will select a set of test images from the dataset.
2. We will apply a square mask (e.g. 24×24 pixels centred on the cat’s face) programmatically zeroing out the masked region.
3. Reconstruct the masked region using the generative model.

For the **Tiny DDPM**, we plan to implement repaint-style inpainting [5]: during the reverse diffusion process, at each timestep t the known (unmasked) pixels are replaced by forward-diffusing the original image to step t , while the masked region is denoised freely by the model. The reduced $T = 200$ schedule makes this approach approximately 5 times faster per image compared to the standard $T = 1000$ DDPM, which is especially relevant for inpainting where the reverse process must be run in full.

For comparison, we also test a simpler β -VAE-based inpainting approach: encode the masked image, decode from the resulting latent code, and blend the decoded output into the masked region. We compare the two approaches qualitatively and, where feasible, quantitatively via per-pixel MSE and perceptual similarity (LPIPS) on the masked region.

2.6 Style Transfer

We plan to implement neural style transfer applied to cat images generated by one of our models, which will be further selected. The procedure follows Gatys et al. [6]: given a content image and a style image, we optimise a target image to minimise a combined loss:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{content}} + \gamma \mathcal{L}_{\text{style}}$$

where $\mathcal{L}_{\text{content}}$ is the MSE between intermediate feature maps of the content and target images (extracted from a pre-trained VGG-19 network), and $\mathcal{L}_{\text{style}}$ is computed via Gram matrix matching across multiple layers. We will experiment with at least two different artistic styles. Results will be presented qualitatively with discussion of how well the style is transferred while preserving recognisable cat features.

2.7 Cats and Dogs Extension

We retrain the best-performing model (selected based on FID on the cats-only dataset) on the combined Cats and Dogs dataset. We generate a batch of samples and evaluate:

- Whether the generated images clearly resemble either cats or dogs, or produce hybrid artefacts.
- How FID changes compared to the cats-only model.
- Whether class-conditional generation emerges naturally or whether explicit conditioning would be required.

2.8 Mode Collapse Mitigation

If mode collapse is observed during GAN training (detected by visual inspection of generated samples and reduced diversity in pixel-level statistics), we will apply one or more of the following strategies:

- Spectral normalisation of the discriminator.
- One-sided label smoothing (real labels set to 0.9 instead of 1.0).
- Reduced discriminator-to-generator training ratio.

3 Experimental Setup

3.1 General Approach

Our experiments follow a two-phase structure. In the first phase, we train all three models with their baseline hyperparameter configurations and evaluate generation quality. In the second phase, we perform sequential hyperparameter tuning for each model (one hyperparameter at a time, fixing the other at its current best value) and select the best configuration based on FID.

Latent interpolation, inpainting, style transfer, and the cats-and-dogs extension are conducted using the best model checkpoints from the second phase.

It is assumed that each of the considered method will have different number of training epochs so as to handle their different characteristics and structures. For the DCGAN network this number will be between 120-150 epochs, the Tiny DDPM will have 100-150 training iterations, while for the β -VAE 80-100 cycles will be performed. However, these values are only hypothetical and may be changed if the learning procedure will last longer than it is planned.

3.2 Evaluation Metrics

Fréchet Inception Distance (FID) will serve as the primary quantitative metric for generation quality. For each model configuration, we generate images and compute FID against the real training images using the `pytorch-fid` library.

Qualitative assessment is performed by visual inspection of generated samples arranged in grids, evaluating sharpness, diversity, and presence of artefacts.

Inpainting quality is assessed via MSE and LPIPS computed on the masked region, alongside qualitative comparison between the Tiny DDPM and β -VAE approaches.

Training dynamics are monitored via generator and discriminator loss curves (DCGAN), denoising loss curves (Tiny DDPM), and ELBO decomposed into reconstruction and β -weighted KL terms (β -VAE). FID is tracked at regular intervals during training to detect instabilities and confirm convergence.

3.3 Reproducibility

To ensure reliability and transparency, all random number generators (weight initialisation, data shuffling, noise sampling) are initialised with a constant set of three different seeds. Every experiment is executed three times, and we will report the mean and standard deviation of FID scores.

References

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